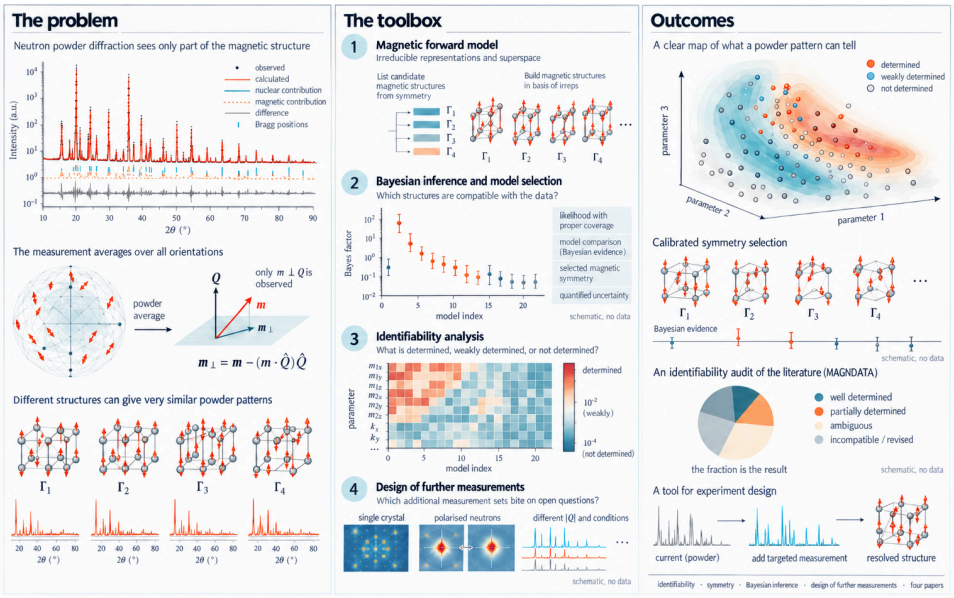
What a powder pattern can tell

Identifiability, Bayesian inference and symmetry in the determination of magnetic structures

ABSTRACT

More than two thousand magnetic structures are published, most from neutron powder diffraction, and the database collecting them reports that many were described ambiguously; even NiO and MnO were revised in 2024. The measurement is the reason: the neutron sees only the component of the moment perpendicular to the scattering vector, and the powder average discards orientation, so part of a structure is not in the data, part only weakly, and part of what is reported is the model's choice. Symmetry tools list the candidates; the choice is still made by agreement factors and judgement. This thesis asks which features of a magnetic structure a powder pattern determines, which it does not, and what cuts the rest, for one pattern and for the literature. Four contributions: a magnetic forward model in irreducible-representation coordinates, in superspace, that prints the directions the data do and do not fix; magnetic symmetry chosen by Bayesian evidence with a likelihood whose coverage is tested; an identifiability audit of the commensurate structures of MAGNDATA; and which further measurement settles what the powder leaves open. Expected outcomes: a diagnostic in an open engine, calibrated symmetry selection on the group's data, the identifiability map of the literature, a design tool for beam time, four papers.



Graphical abstract. Every plot is schematic and says so; the fraction in the audit is the result, not a value assumed.

Related tools. diff-ref-struct: jmigartua.github.io/diff-ref-struct-app · Ispilu: jmigartua.github.io/ispilu

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Chemistry, Materials Engineering or a related field, and a Master's degree in Physics, Materials Science, Nanotechnology or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in solid-state physics, magnetism, crystallography or diffraction; having used a refinement program (FullProf, GSAS-II or Jana), or some statistics; some programming, ideally in Python.
- Curiosity about what a measurement can and cannot tell, interest in experiments at large neutron facilities, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

SUPERVISORS

Irene Urcelay-Olabarria and Josu M. Igartua
thermomat research group, Fisika Saila, EHU

CONTACT

Prof. Gabriel A. López
gabrielalejandro.lopez@ehu.eus

this offer online

Characterization, modelling and design of laser-textured surfaces with selective radiative properties

From processing to spectral selectivity: surface descriptors, simulation and physics-informed machine learning

ABSTRACT

The radiative properties of materials govern heat transfer by radiation, and their spectral selectivity is key for emerging technologies such as concentrated solar power (CSP) and radiative cooling. Direct laser interference patterning (DLIP) can produce spectrally selective surfaces, but their optical response is difficult to predict. It depends not only on the pattern geometry but also on the surface state left by the process, such as oxide layers. This project hypothesizes that the radiative behavior of DLIP-textured surfaces can be predicted from a reduced set of geometric and compositional descriptors. To test this, samples will be fabricated under varying processing conditions and characterized using the HAIRL emissometer, UV-VIS-NIR-MIR spectroscopy, and surface and compositional analysis. These measurements will link processing parameters to surface descriptors, and those descriptors to radiative properties. Rigorous coupled-wave analysis (RCWA) simulations that include the measured oxide layers will be combined with experimental data to train a physics-informed machine learning model. This addresses the gap between idealized simulations and real surfaces. Finally, the model will be validated by designing, fabricating, and measuring surfaces with a target spectral response. The expected outcome is a validated framework and a set of design rules for tailoring the spectral selectivity of laser-textured surfaces.

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Materials Engineering or a related field, and a Master's degree in Materials Science, Physics, Nanotechnology or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in optics and radiative heat transfer, or in materials characterization techniques (spectroscopy, microscopy or surface analysis); machine learning or numerical simulation (e.g., electromagnetic methods such as RCWA or FDTD); some programming, ideally in Python, for data analysis.
- Interest in experimental laboratory work and in linking measurement, simulation and design, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

SUPERVISORS

Josu M. Igartua and Jon Gabirondo-López
thermomat research group, Fisika Saila, EHU

CONTACT

Prof. Gabriel A. López
gabrielalejandro.lopez@ehu.eus

this offer online

Development of an emissometer for the study of ceramic materials at very high temperatures for energy efficiency processes

Laser heating, calibration and uncertainty of spectral emissivity up to 2000 °C

ABSTRACT

The study of the optical-radiative properties of ceramic materials at very high temperatures (T > 1000 °C) is an area of great technological interest. Understanding these properties allows researchers to evaluate energy efficiency and remotely determine the temperature of materials as diverse as thermal radiators intended to replace flames in industrial electrification processes or high-temperature thermal storage systems.

The project seeks to develop an experimental device that enables the study of the infrared emissivity of ceramic materials at temperatures up to 2000 °C using laser heating (Figure 1), alongside measurement and calibration methods to calculate spectral emissivity and estimate its uncertainty. Once the emissometer is developed, it will be validated using reference materials, such as silicon carbide or alumina, before studying ceramic materials candidates for use as radiators in industrial ovens and thermal storage systems. Finally, the infrared radiation emission mechanisms of the ceramics will be studied using various optical models to interpret the results and determine which materials possess the appropriate properties for these technologies.

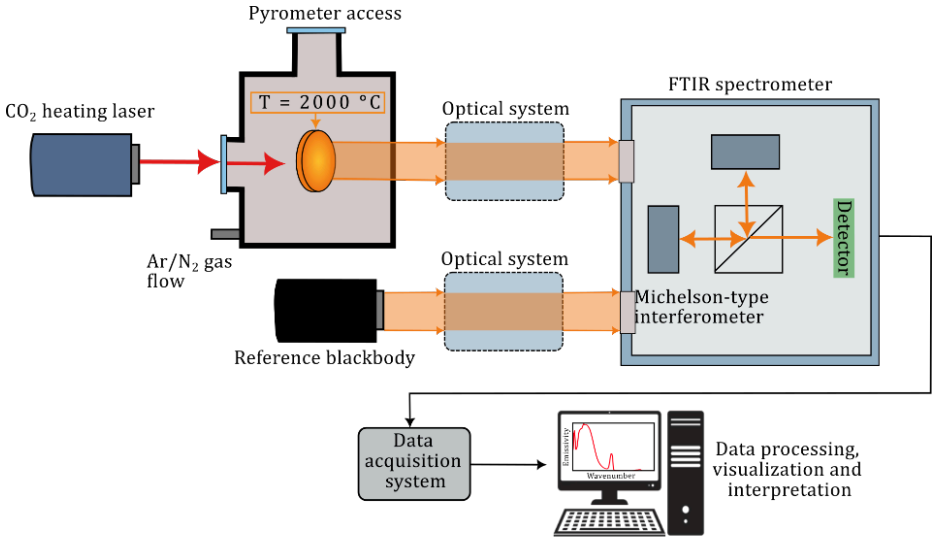


Figure 1. Proposed experimental configuration for spectral emissivity measurements up to 2000 °C

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Electronic Engineering, Industrial Engineering, the Dual Degree in Physics and Electronic Engineering or a related field, and a Master's degree in Materials Science, Electronic Engineering, Mechanical Engineering, Industrial Engineering or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in instrumentation, automatic control or experimental physics, and in infrared optics or measurement uncertainty; some programming, ideally in Python.
- Interest in building, calibrating and validating an experimental device, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

SUPERVISORS

Mireia Sainz-Menchón and Telmo Echániz
thermomat research group, Fisika Saila, EHU

CONTACT

Prof. Gabriel A. López
gabrielalejandro.lopez@ehu.eus

this offer online

Advanced Joining of High-Performance Metallic Materials

Transient liquid phase bonding of nickel-based superalloys to stainless steel, from processing to properties

ABSTRACT

Are you interested in materials science, physical metallurgy, and advanced microscopy? Join our research team at the University of the Basque Country (UPV/EHU), Spain, and contribute to cutting-edge research on metallic materials for demanding industrial applications.

This PhD project focuses on the development and characterization of dissimilar joints between nickel-based superalloys, such as Inconel 718 and Haynes alloys, and stainless steel, using Transient Liquid Phase Bonding (TLPB). This joining technique offers exciting opportunities to combine the excellent high-temperature performance of superalloys with the cost-effectiveness of stainless steel, particularly for applications in energy generation, aerospace, and corrosive environments.

The research will investigate how processing conditions influence microstructure, mechanical performance, and corrosion resistance. You will gain hands-on experience with advanced characterization techniques, including scanning and transmission electron microscopy (SEM/TEM), electron backscatter diffraction (EBSD), energy-dispersive X-ray spectroscopy (EDS), and focused ion beam (FIB) microscopy, alongside mechanical and electrochemical testing.

The project is supported by an international collaboration with research groups in Austria and Argentina, offering opportunities for international research stays and interdisciplinary teamwork.

Join us to explore the fascinating relationship between processing, microstructure, and properties, and develop the experimental and analytical skills needed for a career in advanced materials research.

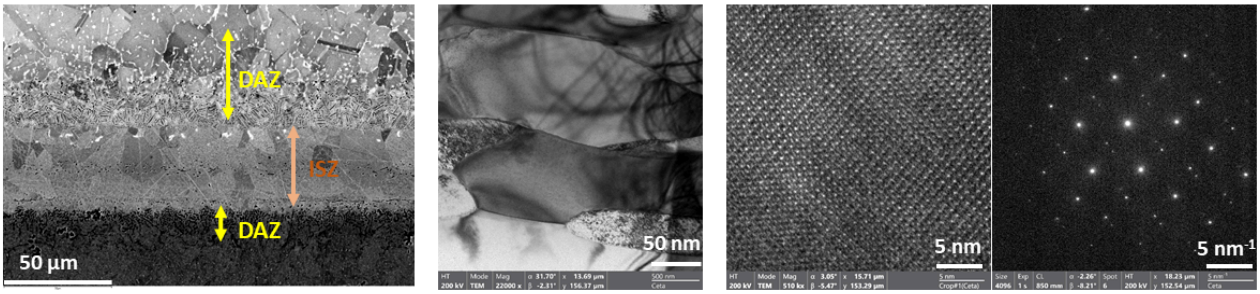


Figure 1. Multiscale characterization of the dissimilar join using scanning and transmission electron microscopy (SEM and TEM), from the micrometre to the nanometre scale. The PhD project will involve advanced microstructural characterization to investigate the relationships between processing conditions, microstructure, and the mechanical and corrosion properties of dissimilar metallic joints.

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Chemistry or Engineering, and a Master's degree in Materials Science and Engineering or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in Materials Science and Engineering or Mechanical Properties of Materials; previous experience in electron microscopy and X-ray diffraction (XRD); some programming, ideally in Python, for data analysis.
- Motivation to pursue experimental research, interest in international research stays (Austria and Argentina), willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

SUPERVISORS

Iñaki López-Ferreño and Gabriel A. López
thermomat research group, Fisika Saila, EHU

CONTACT

Prof. Gabriel A. López
gabrielalejandro.lopez@ehu.eus

this offer online